

DEPENDENCY_MAP

Helios — Artifact Dependency Map & Build Order

Companion to: HELIOS_PROJECT_BIBLE.md · Version: v1.0 · Date: 2026-06-03

Purpose. This document is the build-time complement to the Bible. The Bible says *what* every artifact is; this map says *what must exist before each artifact can be built, in what order to build them, which artifacts are load-bearing, and what to build first*. It is optimized for **long-term project continuity**: an engineer returning years later (or an AI agent resuming the build) should be able to read this map plus the Bible and reconstruct Helios in dependency-safe order without rediscovering the graph.

Every artifact below traces to a Bible section (cited as §). The only artifact that currently exists is the Bible itself and this map; everything else is "remaining."

0. Legend & conventions

- **Tier (T0–T13):** topological layers. An artifact may only depend on its own tier or lower tiers. No back-edges = acyclic by construction.
- **depends on :** *direct* upstream artifacts (transitive deps omitted for readability).
- **★ Keystone:** load-bearing artifact — many things break or are wrong if this is wrong. Build with the most care and the most tests.
- **|| Parallelizable:** sits on an independent subgraph and can be built concurrently with the main data spine.
- **Maturity tag:** [P0] MVP/L1, [P1] Strong Portfolio/L2, [P2] Top-1%/L3, [P3/P4] production/frontier (§23).

1. Full artifact inventory (tiered, with direct dependencies)

T0 — Repository & environment foundation [P0]

ID	Artifact	Depends on	Bible	Notes
A0.1	Repo scaffold (<code>pyproject.toml</code> / <code>requirements.txt</code> , <code>.gitignore</code> , package layout <code>helios/</code>)	—	§16.1	Python 3.11; deps: google-cloud-bigquery, scipy, statsmodels, prophet/pmdarima, mcp, anthropic.
A0.2	CLAUDE.md ★	Bible	§12 (workflows)	Continuity anchor for AI-assisted dev: canonical names, grounding rules G1–G5, dbt conventions.
A0.3	GCP project + service account + IAM (<code>bigquery.dataViewer</code> , <code>bigquery.jobUser</code>) + ADC	—	§17, §18.3	Read-only on <code>bigquery-public-data</code> . Prereq for any query.
A0.4	<code>dbt_project.yml</code> , <code>profiles.yml</code> , <code>packages.yml</code>	A0.1, A0.3	§16.7–16.8	Materializations, partition/cluster config; targets <code>helios_dev</code> / <code>helios_prod</code> .
A0.5	<code>mcp_servers.yaml</code> (stub → finalized)	A0.1	§18.10	Stub at T0; finalize per server as built (T6).

T1 — dbt sources, macros, seeds [P0] (depends: T0)

ID	Artifact	Depends on	Bible	Notes
A1.1	<code>macros/get_event_param.sql</code>	A0.4	§16.6	Typed <code>event_params</code> extraction; used by all staging.
A1.2	<code>macros/sessionize.sql</code>	A1.1	§16.6	Builds <code>(user_pseudo_id, ga_session_id)</code> key.
A1.3	<code>macros/channel_group.sql</code> (<code>channel_group_case()</code>) ★	A0.4	§16.6 / §12.5	Single source of truth for the 10 channel groups.

ID	Artifact	Depends on	Bible	Notes
A1.4	<code>macros/test_revenue_reconciles.sql</code> (custom generic test)	A0.4	\$16.5	Underpins reconciliation guarantee.
A1.5	<code>seeds/channel_group_mapping.csv</code>	A0.4	\$16.1	source/medium → channel_group seed.
A1.6	<code>models/staging/src_ga4.yml</code> (source decl + freshness)	A0.4	\$16.3	Declares <code>events_*</code> shards.

T2 — Staging models [P0] (depends: T1)

ID	Artifact	Depends on	Bible	Notes
A2.1	<code>stg_ga4_events.sql</code>	A1.1, A1.2, A1.6	\$16.9	1:1 typed/renamed events; surfaces <code>session_key</code> , source/medium, revenue.
A2.2	<code>stg_ga4_event_params.sql</code>	A1.6	\$8.1	Unnested long key/value table.
A2.3	<code>stg_ga4_schema.yml</code> (tests + docs)	A2.1, A2.2	\$16.1	not_null/unique on keys.

T3 — Intermediate models [P0] — THE DATA KEYSTONE (depends: T2)

ID	Artifact	Depends on	Bible	Notes
A3.1	<code>int_ga4_sessionized.sql</code> ★	A2.1, A2.2, A1.2, A1.3	\$8.5	Session row, <code>landing_page</code> , session-scoped source/medium (traffic_source gotcha), engagement. If wrong, every metric is wrong.
A3.2	<code>int_ga4_funnel_steps.sql</code> ★	A2.1, A3.1	\$10, \$8.3	<code>reached_*</code> max-downstream monotonic flags. Drives the whole funnel.
A3.3	<code>int_ga4_schema.yml</code>	A3.1, A3.2	\$16.1	Funnel-monotonicity & uniqueness tests.

T4 — Marts (facts + dims) [P0/P1] (depends: T3)

ID	Artifact	Depends on	Bible	Notes
A4.1	<code>dim_date.sql</code>	A0.4	\$8.4	Date spine 2020-11-01..2021-01-31; near-independent.
A4.2	<code>dim_channels.sql</code>	A1.3, A1.5	\$8.4	10 channel groups; PK channel_group.
A4.3	<code>dim_users.sql</code>	A3.1	\$8.4 / \$8.6	First-touch attrs; cookie-grain identity.
A4.4	<code>dim_items.sql</code>	A2.1	\$8.4	Item/category catalog from items[].
A4.5	<code>fct_sessions.sql</code>	A3.1, A4.2	\$8.3	Session attributes/engagement.
A4.6	<code>fct_funnel.sql</code> ★	A3.1, A3.2	\$8.3	One row/session + <code>reached_*</code> + session_revenue. Grain-defining session fact.
A4.7	<code>fct_daily_funnel.sql</code> ★	A4.6	\$8.3 / \$16.10	Daily × dimension counts incl. <code>revenue</code> . Primary feed for Monitor + Decompose + eval injector. Pinned to aggregate <code>fct_funnel</code> (A4.6), which carries <code>session_revenue</code> ; the \$16.10 sample reading from <code>int_ga4_funnel_steps</code> is an abbreviated illustration (counts only, no revenue).
A4.8	<code>fct_orders.sql</code>	A2.1	\$8.3	One row/transaction; revenue/gross/net/aov inputs.
A4.9	<code>fct_order_items.sql</code>	A2.1, A4.8, A4.4	\$8.3	Exploded items[].
A4.10	<code>fct_funnel_by_dim.sql</code>	A4.6, A4.1–A4.2	\$16.2	Decomposition input (mix vs rate).
A4.11	<code>fct_cohorts.sql</code>	A4.3, A4.8	\$16.2	Retention input for <code>cohort_retention</code> . [P2]

ID	Artifact	Depends on	Bible	Notes
A4.12	<code>core__/finance__/growth__schema.yml</code> schema tests, <code>snapshots/snap_dim_items.sql</code>	A4.1–A4.11	§16.1, §16.4	Generic tests incl. <code>accepted_values</code> on <code>channel_group</code> , <code>revenue_reconciles</code> . (Singular test = A4.14.)
A4.13	<code>exposures/exposures.yml</code>	A4.* (finalize after T8 — scheduled in M9)	§16.1	Declares agents + Decision Brief as consumers; lineage.
A4.14	<code>tests/assert_session_conversion_rate_bounds.sql</code> (singular data test)	A4.7	§16.1, §15.4	Asserts <code>session_conversion_rate</code> ∈ [0,1] and funnel-rate bounds, computed inline from <code>fct_daily_funnel</code> counts (no semantic-layer dep). Runs in CI.

T5 — Semantic layer [P0/P1] — THE GOVERNANCE KEYSTONE (depends: T4)

ID	Artifact	Depends on	Bible	Notes
A5.1	<code>models/semantic/semantic_layer.yml</code> (metric + dimension registry) ★★	A4.6, A4.7, A4.8	§14.1–14.3	The single source of truth for all governed SQL. Everything that queries goes through this.
A5.2	<code>metrics__schema.yml</code> + referential-integrity compile check	A5.1	§14.4–14.5	Dangling-reference = hard fail; versioned.

T6 — MCP servers [P0→P2] (depends: T5, T4, T0; report-mcp depends on T7)

ID	Artifact	Depends on	Bible	Notes
A6.0	MCP shared scaffolding (FastMCP base, JSON-RPC, schemas)	A0.1	§18.3, §18.12	Common server harness.
A6.1	<code>warehouse-mcp</code> ★	A6.0, A0.3, A4.*	§18.4, §18.12	Sole BigQuery client; <code>dry_run</code> → <code>run_query</code> byte-budget gate; <code>reconcile</code> .
A6.2	<code>semantic-mcp</code> ★	A6.0, A5.1	§18.5, §14.4	The ONLY path to SQL; <code>build_query</code> chokepoint. Output feeds A6.1.
A6.3	<code>stats-mcp</code> ★ II	A6.0	§18.6	The ONLY path to math. <code>decompose_change</code> = thesis centerpiece. Data-independent → build early in parallel.
A6.4	<code>experiment-mcp</code> II	A6.0	§18.7	power/runtime/design; no data access. [P1]
A6.5	<code>report-mcp</code> core (<code>render_brief</code> , <code>export</code>) II	A6.0	§18.8	Stateless rendering/export; no memory dependency → buildable at P0. Memory-backed tools are split out to A7.4.

T7 — Memory store [P1/P2] (depends: T0; gates report-mcp + Critic)

ID	Artifact	Depends on	Bible	Notes
A7.1	<code>helios_memory</code> DDL (diagnosis_history, suppression_list, glossary, seasonality_calendar, launch_calendar, action_tracking, run_state, audit_log) ★	A0.3	§22.1–22.5	Durable system of record.
A7.2	Vector store setup (embeddings of findings)	A7.1	§22.1	ANN recall for <code>recall_prior</code> .
A7.3	Memory seeds (glossary; seasonality: Black Friday 2020 / Dec peak / Jan trough; launch_calendar) II	A7.1	§22.3	Confound priors the Critic needs.
A7.4	<code>report-mcp</code> memory tools (<code>save_diagnosis</code> , <code>recall_prior</code> + vector upsert) ★	A6.5, A7.1, A7.2	§18.8, §22.1	The stateful half of report-mcp; the sole memory-I/O path for agents (resolves the vector store's writer/reader). Depends on lower-tier A6.5 + same-tier A7.1/A7.2 — no back-edge.

T8 — Agents (Claude Agent SDK) [P0→P2] (depends: T6 + T7)

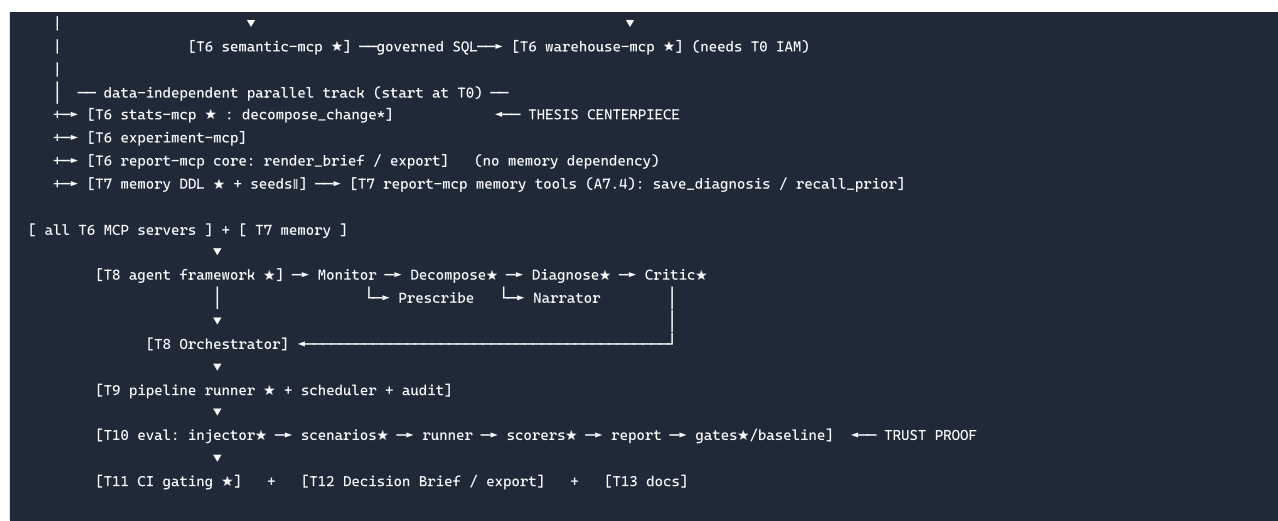
ID	Artifact	Depends on	Bible	Notes
A8.0	Agent framework: control plane / FSM, typed JSON envelope, per-agent allow-lists, context windowing, retries ★	A6.0–A6.3, A6.5	\$18.9, \$19.2, \$19.5	Shared substrate. Scoped to the MCP servers the minimal loop needs (not the memory tools A7.4), so the framework is buildable at P0.
A8.1	Monitor	A8.0, A6.2, A6.1, A6.3	\$19.1	<code>detect_anomaly</code> over canonical series. [P0]
A8.2	Decompose ★	A8.0, A6.2, A6.1, A6.3	\$19.1	<code>decompose_change</code> mix/rate/interaction. [P1]
A8.3	Diagnose ★	A8.0, A6.2, A6.1, A6.3	\$19.4	Hypothesis-tree RCA, SQL-verified. [P1]
A8.4	Critic ★	A8.0, A6.1–A6.3, A7.4	\$19.3	Adversarial refutation; PASS/DOWNGRADE/DROP. Uses <code>recall_prior</code> → depends on A7.4 (which transitively pulls report-mcp + memory). [P1]
A8.5	Prescribe	A8.0, A6.4	\$19.1	Powered experiment backlog. [P1]
A8.6	Narrator	A8.0, A6.5	\$19.1	Renders brief via <code>render_brief</code> (A6.5) for the minimal loop; full loop also calls <code>save_diagnosis</code> → adds A7.4. [P0 minimal]
A8.7	Orchestrator	A8.1–A8.6, A7.4	\$19.3, \$19.6	Plans run, drives FSM, budget, routes to Critic. Calls <code>recall_prior</code> → depends on A7.4 (not raw memory).

T9 — Orchestration runtime [P1/P2] (depends: T8)

ID	Artifact	Depends on	Bible	Notes
A9.1	Pipeline runner (PLAN→MONITOR→...→END FSM) ★	A8.7	\$19.3	Executes a full run; consumed by eval runner.
A9.2	Scheduler endpoint (Cloud Scheduler/cron)	A9.1	\$19.6, \$23.3	Autonomy. [P2]
A9.3	run_state/audit wiring	A9.1, A7.1	\$22.5	Budget enforcement + reconstructability.

T10 — Evaluation harness [P1] — THE TRUST PROOF (depends: T4.7, T9, T6, T5)

ID	Artifact	Depends on	Bible	Notes
A10.1	<code>helios_eval.fct_daily_funnel_base</code> (frozen extract)	A4.7, A6.1	\$20.2	Baseline copy extracted from <code>fct_daily_funnel</code> via <code>warehouse-mcp.run_query</code> ; never mutate the public source. (<code>_perturbed</code> + <code>labels</code> are produced by the injector A10.2, not here.)
A10.2	<code>eval/injector.py</code> ★	A10.1, A6.1	\$20.2	Seeded rate/volume perturbation + ground-truth labels.
A10.3	<code>eval/scenarios/*.yaml</code> (50) ★	A10.2	\$20.3	Coverage buckets incl. seasonality decoys + controls.



No edge points upward → the graph is acyclic; any topological sort is a valid build order.

3. Recommended generation order (phased, topologically safe)

Build in **milestones**; never start a milestone until the prior one is green (dbt build + tests, and from M7 onward the eval harness). Mapped to Bible §23 phases.

#	Milestone	Artifacts	Gate to advance
M0	Foundation [P0]	A0.1–A0.5 (repo, GCP/IAM, dbt cfg, CLAUDE.md, mcp stub)	dbt debug connects; ADC authenticates.
M1	dbt sources & macros [P0]	A1.1–A1.6	macros compile; seed loads.
M2	Staging [P0]	A2.1–A2.3	staging tests pass; row counts sane.
M3	Sessionization & funnel flags [P0] ★	A3.1, A3.2, A3.3	funnel monotonicity test passes; session_key unique.
M4	Core + finance marts [P0]	A4.1–A4.9, A4.12	revenue_reconciles to the cent (§23.1 exit); channel accepted_values passes.
M5	Semantic registry [P0] ★★	A5.1, A5.2	registry compiles; 0 dangling references.
M6	Grounding MCP pair [P0]	A6.0, A6.2 (semantic), A6.1 (warehouse)	build_query→dry_run→run_query round-trips; budget gate fires; reconcile matches.
M6b	Stats/experiment/report MCP [P0/P1]	A6.3 (stats★), A6.4, A6.5 (report-mcp core)	decompose_change unit tests vs hand-worked mix/rate/interaction examples pass; render_brief emits a brief.
M7	Minimal loop [P0] (L1 exit)	A8.0 (framework, scoped to A6.0–A6.3/A6.5), A8.1 (Monitor), A8.6 (minimal Narrator via render_brief), A12.1 (basic brief)	one anomaly → brief in <5 min; 0 hallucinated columns. L1 DONE . (All deps — incl. report-mcp core A6.5 — exist from M6/M6b.)
M8	Memory + report-mcp memory tools [P1]	A7.1–A7.3, A7.4	save/recall round-trips; seeds loaded; vector ANN recall works.
M9	Full agent loop [P1]	A8.2 Decompose, A8.3 Diagnose, A8.4 Critic, A8.5 Prescribe, A8.7 Orchestrator, A9.1 runner, A9.3 audit, A4.13 exposures (finalize)	a real run produces PASS findings with significance + \$ + action.
M10	Eval harness + CI [P1] ★ (L2 exit)	A10.1–A10.7, A11.1	≥85% top-1 vs ≤45% baseline; hallucination 0%; cost under budget. L2 DONE .

#	Milestone	Artifacts	Gate to advance
M11	Autonomy & depth [P2] (L3 exit)	A9.2 scheduler, A4.11 cohorts, forecast / cohort_retention / rfm_segment usage, full Critic battery, A12.2/A12.3	<5 min/run on schedule; accuracy holds across all dims. L3 DONE.
M12+	Productionization / frontier [P3/P4]	per §23.4–23.5	warehouse-agnostic adapters; causal inference; auto-executed experiments.

4. Critical path (the spine)

The longest must-be-sequential chain — and therefore the schedule-determining path — to the **L2 trust milestone (the 85%-vs-45% claim)**:

```
T0 foundation → A1.* macros → A2.1 staging → A3.1 sessionized → A3.2 funnel_steps
→ A4.6 fct_funnel → A4.7 fct_daily_funnel → A5.1 semantic registry
→ A6.2 semantic-mcp (+ A6.1 warehouse-mcp) → A6.3 stats-mcp/decompose_change
→ A8.0 agent framework → A8.1→A8.2→A8.3→A8.4 (Monitor→Decompose→Diagnose→Critic) → A8.7 Orchestrator
→ A9.1 pipeline runner → A10.2 injector → A10.3 scenarios → A10.4 runner → A10.5 scorers → A10.7 gates → A11.1 CI
```

Co-critical branch (must converge before A10.4). Because the eval runner executes *all seven* agents, the Critic's `recall_prior` (A8.4) and the Narrator's `save_diagnosis` (A8.6) require the memory branch:

```
A7.1 memory DDL → A7.2 vector store + A7.3 seeds → A7.4 report-mcp memory tools
→ consumed by A8.4 Critic (recall_prior) and A8.6 Narrator (save_diagnosis) → A10.4 full-pipeline run
```

This branch runs in parallel with the spine but is *co-critical from A8.4 onward* — the full 7-agent eval cannot run without it.

Anything **not** on the spine *or* the co-critical branch (finance facts, cohorts, experiment-mcp, the scheduler, drill-down UI, docs) can be built in parallel or deferred without delaying the headline claim.

5. Most important artifacts (ranked, with why)

1. **A5.1 semantic_layer.yaml registry ★★** — the governance keystone. Every governed query, every metric, `semantic-mcp`, and the hallucination-rate guarantee derive from it. For continuity, this YAML + the Bible is enough to regenerate all SQL. **Highest leverage.**
2. **A3.1/A3.2 sessionization + reached_* flags ★** — the data keystone. If sessionization or the monotonic funnel flags are wrong, *every* downstream number is silently wrong. Maximum test coverage here.
3. **A6.3 stats-mcp / decompose_change ★** — the thesis centerpiece (mix vs rate vs interaction, Simpson's-paradox defense). Data-independent, so build and unit-test it early against hand-worked examples; it is also the single most interview-defensible component.
4. **A10. eval harness + scenarios + gates ★*** — the trust proof and the *continuity* guard: the CI gate is what lets a future engineer refactor without silently regressing the 85% claim. This is what separates Helios from a demo.
5. **A6.1/A6.2 warehouse-mcp + semantic-mcp ★** — the grounding boundary made physically enforceable (only path to SQL; mandatory dry-run/byte budget).
6. **A0.2 CLAUDE.md + A13.2 this map + the Bible** — the human/AI continuity anchors. Without them the graph must be rediscovered.
7. **A8.0 agent framework + A8.4 Critic ★** — the control plane and the adversarial verifier; the Critic is why findings are trustworthy enough to ship autonomously.

6. What to generate FIRST (the immediate next 6 steps)

These unblock the widest fan-out with the least prerequisite. Do them in this order:

1. **A0.1–A0.4** — repo scaffold, GCP/IAM/ADC, `dbt_project.yml / profiles.yml / packages.yml`, and **CLAUDE.md** (so all subsequent AI-assisted work is grounded).
2. **A1.1–A1.6** — the four macros (esp. `channel_group_case` ★), source declaration, and seed.
3. **A2.1–A2.3** — staging models + tests.
4. **A3.1 → A3.2** — sessionization, then `reached_*` funnel flags. (*Spend disproportionate care + tests here.*)
5. **A5.1** — the semantic registry, immediately after the funnel facts (A4.6/A4.7) exist. (*This is the governance keystone — front-load it.*)

6. **A6.3 stats-mcp in parallel** from day one (data-independent): implement `decompose_change` with unit tests before any agent needs it.

Continuity rule of thumb: front-load the two keystones (A3.* sessionization, A5.1 registry) and the data-independent thesis core (A6.3), because they have the largest downstream blast radius and are the artifacts most expensive to get wrong late.

7. Parallelizable tracks (independent subgraphs)

To compress wall-clock without violating the DAG, three tracks can run concurrently after T0:

- **Track A — Data spine (critical path):** T1 → T2 → T3 → T4(core) → T5 → T6(semantic+warehouse).
- **Track B — Math & experiment (data-independent):** A6.3 `stats-mcp` (incl. `decompose_change`, `forecast`, `cohort_retention`, `rfm_segment`) + A6.4 `experiment-mcp`, each with their own unit tests. Joins Track A at T8.
- **Track C — Memory & finance (independent):** A7.1–A7.3 memory DDL/seeds + A4.8/A4.9 finance facts + A4.1 `dim_date` / A4.4 `dim_items`. Track C gates `report-mcp` (A6.5) and cohorts (A4.11).

All three converge at **T8 (agents)**, which is the first artifact requiring the full stack.

8. Continuity guardrails (do not skip)

- **Never let the registry (A5.1) drift from the marts (T4).** A column rename in a fact must update exactly one YAML; CI referential-integrity (A5.2) enforces it.
- **The CI eval gate (A11.1) is the regression firewall.** Once M10 is green, no change merges that drops top-1 accuracy >2pts or introduces any hallucination. This is the single most important continuity mechanism.
- **Keep CLAUDE.md and this map updated as artifacts land** — they are the resume points for any future build session.
- **Seed memory (A7.3) before relying on the Critic's seasonality refutations** — an unseeded `seasonality_calendar` makes the Critic blind to known confounds (e.g., the Black Friday 2020 / January-trough swings in this dataset window).
- **Honor the keystones' test budgets:** sessionization, funnel monotonicity, revenue reconciliation, and `decompose_change` algebra each need golden-value tests, because they fail *silently* (wrong numbers, not errors).
- **Keep the registry filename canonical.** The semantic registry is `models/semantic/semantic_layer.yaml` (the v2 registry; the retired v1 `semantic_models.yaml` is archived in `docs/archive/superseded/`); `semantic-mcp` in `mcp_servers.yaml` must point its `registry:` at that exact path. A5.1 and A6.2 share one physical file — a divergence here silently desyncs the governed-SQL guarantee.
- **Pin the `fct_daily_funnel` source.** It aggregates `fct_funnel` (A4.6) — the source carrying `session_revenue` — not `int_ga4_funnel_steps`; the \$16.10 sample is an abbreviated counts-only illustration. Getting this wrong drops `revenue` from the daily grain and breaks the eval injector's dollar-at-risk labels.